

Multi-Model Speaker Identification Report

Generated 09-08-2026 11:48:17 AM - 4 trained models evaluated

Audio Details

Field	Value
Filename	692.wav
Duration	1.000 seconds
Sample Rate	16000 Hz
Channels	1
Bit Depth / Subtype	PCM_16
Format	WAV
File Size	31.29 KB
Peak Amplitude	0.727997
RMS Energy	0.199504

Model Comparison Summary

Model	Predicted Speaker	Confidence	Inference
SpeechBrain - ECAPA-TDNN	Nelson_Mandela	97.94%	71.56 ms
SpeechBrain - X-Vector	Nelson_Mandela	80.37%	504.23 ms
Microsoft - WavLM Base Plus SV	Nelson_Mandela	89.43%	14266.41 ms
Microsoft - UniSpeech-SAT Base Plus SV	Nelson_Mandela	87.15%	3341.22 ms

Model 1: SpeechBrain - ECAPA-TDNN

Model Details

Field	Value
Model ID	speechbrain_ecapa
Provider	SpeechBrain
Architecture	ECAPA-TDNN
Pretrained Source	speechbrain/spkrec-ecapa-voxceleb
Embedding Dimension	192
Classifier	Calibrated RBF Support Vector Machine

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Confidence	97.94%
Inference Time	71.56 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	97.94%
2	Julia_Gillard	1.17%
3	Jens_Stoltenberg	0.43%
4	Magaret_Tarcher	0.35%
5	Benjamin_Netanyau	0.12%

Related Audio Files

No.	Audio filename
1	331.wav
2	332.wav
3	333.wav
4	334.wav
5	335.wav

Model 2: SpeechBrain - X-Vector

Model Details

Field	Value
Model ID	speechbrain_xvector
Provider	SpeechBrain
Architecture	X-Vector TDNN
Pretrained Source	speechbrain/spkrec-xvect-voxceleb
Embedding Dimension	512
Classifier	Calibrated RBF Support Vector Machine

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Confidence	80.37%
Inference Time	504.23 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	80.37%
2	Magaret_Tarcher	19.59%
3	Jens_Stoltenberg	0.04%
4	Julia_Gillard	0.00%
5	Benjamin_Netanyau	0.00%

Related Audio Files

No.	Audio filename
1	331.wav
2	332.wav
3	333.wav
4	334.wav
5	335.wav

Model 3: Microsoft - WavLM Base Plus SV

Model Details

Field	Value
Model ID	wavlm_base_plus_sv
Provider	Microsoft
Architecture	WavLM Transformer
Pretrained Source	microsoft/wavlm-base-plus-sv
Embedding Dimension	512
Classifier	Calibrated RBF Support Vector Machine

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Confidence	89.43%
Inference Time	14266.41 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	89.43%
2	Benjamin_Netanyau	9.96%
3	Jens_Stoltenberg	0.61%
4	Magaret_Tarcher	0.00%
5	Julia_Gillard	0.00%

Related Audio Files

No.	Audio filename
1	331.wav
2	332.wav
3	333.wav
4	334.wav
5	335.wav

Model 4: Microsoft - UniSpeech-SAT Base Plus SV

Model Details

Field	Value
Model ID	unispeech_sat_base_plus_sv
Provider	Microsoft / Hugging Face
Architecture	UniSpeech-SAT + X-Vector Head
Pretrained Source	microsoft/unispeech-sat-base-plus-sv
Embedding Dimension	512
Classifier	Calibrated RBF Support Vector Machine

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Confidence	87.15%
Inference Time	3341.22 ms

Speaker Probability Distribution

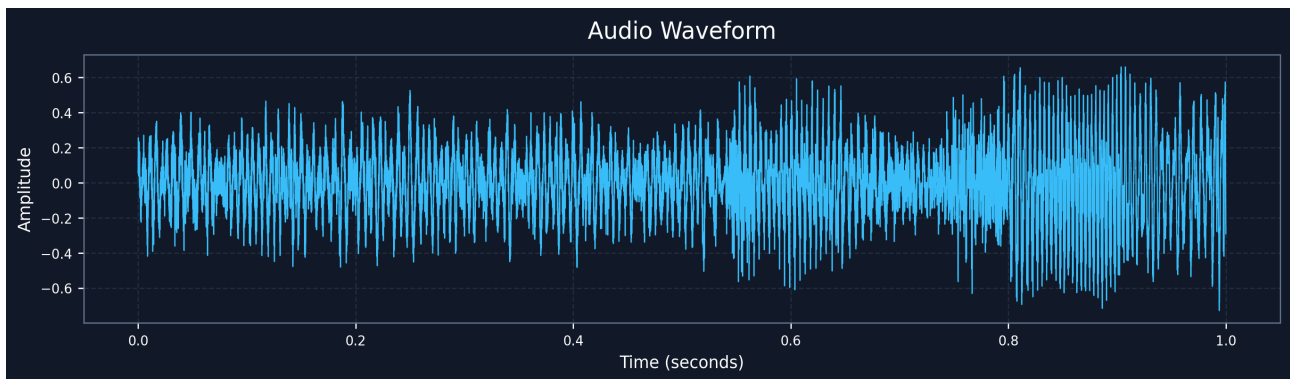
Rank	Speaker	Probability
1	Nelson_Mandela	87.15%
2	Jens_Stoltenberg	12.28%
3	Benjamin_Netanyau	0.57%
4	Julia_Gillard	0.01%
5	Magaret_Tarcher	0.00%

Related Audio Files

No.	Audio filename
1	331.wav
2	332.wav
3	333.wav
4	334.wav
5	335.wav

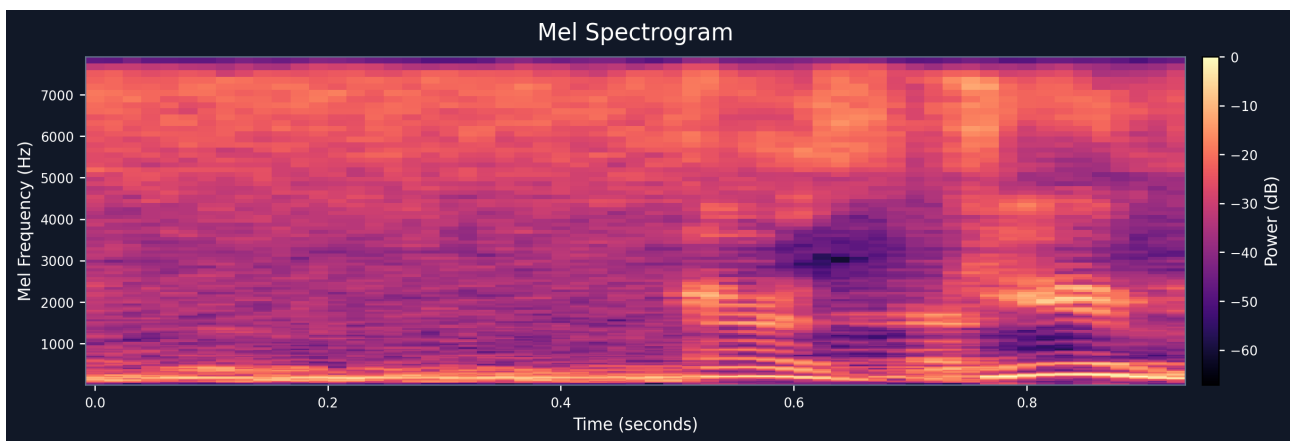
Audio Signal Visualisations

Waveform



Signal amplitude over time. Peak: 0.727997; RMS: 0.199504.

Mel Spectrogram



Acoustic energy across time and Mel-scaled frequency. Colormap: magma.